Lending Club Credit Scoring

P2P Credit Risk Modeling: Temporal Split, Leakage Prevention, and Operational Cut-off Optimization

Course: Statistical Learning – **Group 6**

National Economics University

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Outline

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Project Overview & Data Setup

Member	Student ID	Assigned Tasks	Contribution
Phạm Thùy Anh	11247137	Data pipeline, leakage control, EDA, Lasso.	33.3%
Đinh Vân Anh	11247123	Logistic, Naive Bayes, Decision Tree, Random Forest.	33.3%
Hoàng Thị Lệ Xuân	11247249	KNN, SVM, Metrics, Cut-off optimization, PSI.	33.3%

Core Objective

Predict loan default (*Charged Off* vs. *Fully Paid*) using strictly **application-time information** to reflect real-world credit risk.

Project Overview & Data Setup

Dataset Scope (Lending Club 2007–2018Q4):

- Raw Scale: **2,260,701 rows** $\rightarrow$ Resolved Sample: **1,345,310 loans**.
- Target Definition:
  - **y** $=$ **0** (**Fully Paid**): Repaid in full.
  - **y** $=$ **1** (**Charged Off**): Defaulted.
- Class Imbalance: **4.01 : 1** (20.0% default rate).

Status	Count	%
Fully Paid (0)	1,076,751	80.0%
Charged Off (1)	268,559	20.0%
Total	**1,345,310**	**100%**

Project Overview & Data Setup

Application Timestamp Principle (Step 1 Audit)

Model uses strictly information available at Step 1 (**Application Time**). Items from Step 2 (Underwriting) to Step 6 (Servicing/Recovery) are excluded to prevent target leakage.

Four Primary Post-Origination Excluded Categories:

1. **Repayment & Balances (`repayment_balance`):** `total_pymnt`, `total_rec_prncp`, `total_rec_int`, `out_prncp`, `last_pymnt_d`, `next_pymnt_d` – Servicing logs.
2. **Recovery Metrics (`recovery`):** `recoveries`, `collection_recovery_fee` – Direct label proxies (> 0 strictly post default).
3. **Subsequent Credit Pulls (`later_credit_pull`):** `last_fico_range_high/low`, `last_credit_pull_d` – Updated long post origination.
4. **Distress & Settlement (`hardship`/`settlement`):** `pymnt_plan`, `hardship_flag`, `debt_settlement_flag` – Post-issuance distress.

Project Overview & Data Setup

1. **LC Underwriting Process Outputs (Step 2–4 Exclusions):**
   - **Internal Scoring Outputs:** `grade`, `sub_grade`, `int_rate`, `installment` – LC underwriting outputs. Including them causes the ML model to memorize LC ratings rather than raw risk. Saved as `lc_benchmark`.
   - **Verification Status (`verification_status`):** Excluded to avoid LC process bias (reserved for ablation).
   - **Listing Details:** `funded_amnt`, `initial_list_status`, `disbursement_method` – Listing/funding outputs. Replaced by requested `loan_amnt`.
2. **Policy & Identifier Exclusions:**
   - **Identifiers & Unstructured Text:** `id`, `member_id`, `url`, `desc`, `title`, `emp_title`.
   - **Fair Lending Policy:** `zip_code` removed to prevent proxy discrimination; state location (`addr_state`) kept for macro-regional shock auditing.

Project Overview & Data Setup

1. **Confirmed Retained Application-Time Categories:**
   - **Basic Application:** `loan_amnt`, `term`, `purpose`, `annual_inc`, `dti`, `emp_length`, `home_ownership`, `fico_range_low/high`.
   - **Extended Bureau Data:** `mths_since_*`, `mo_sin_*`, `num_*`, `open_acc`, `revol_util`, `tot_coll_amt`.
   - **Credit File Snapshot (`browseNotes`):** `acc_now_delinq`, `delinq_amnt`, `collections_12_mths_ex_med`, `num_accts_ever_120_pd` – Initial credit pull snapshot at `creditPullD`.
   - **Historical Delinquency (`delinq_2yrs`):** Prior bureau history before application (valid risk factor).

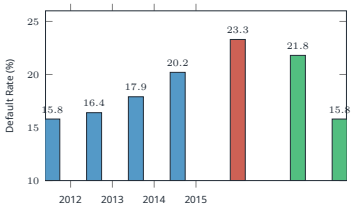
Reverse Verification via Anchored Regex Patterns

Anchored regex keywords (`^total_rec_`, `^last_`, `^next_`, `recover`, `hardship`, `settlement`) verified clean features (e.g., avoiding false catches on `pub_rec_bankruptcies` or `inq_last_6mths`).

Project Overview & Data Setup

Chronological Partitioning (Data Drift Control)

Prevented future-to-past leakage by splitting data strictly by issue date (`issue_d`).



- **TRAIN (2007–2015):** 826,604 loans (61.4%) Def Rate: 18.4%
- **VALIDATION (2016):** 293,095 loans (21.8%) Def Rate: 23.3%
- **TEST (2017–2018):** 225,611 loans (16.8%) Def Rate: 21.3%

Insight: Peak default rate in 2016 highlights macro credit risk shifts, proving why Out-of-Time (OOT) evaluation is essential over random K-Fold CV.

Methodology & Preprocessing

Temporal & Ordinal Transformation

Raw textual fields were converted into strict numerical representations anchored to the application timestamp.

- **Credit History Length (`credit_history_months`):**

$$\text{credit_history_months} = \text{issue_d} - \text{earliest_cr_line}$$

  Calculated strictly relative to loan disbursement date to eliminate look-ahead bias.

- **Employment Length (`emp_length`):** Converted mapped strings (< 1 year $\rightarrow 0$, ..., $10+$ years $\rightarrow 10$).

- **Semantic Missingness Handling:** Missing values in `emp_length` were **kept as NaN** rather than imputed as 0. Unemployed status differs fundamentally from undisclosed employment.

Methodology & Preprocessing

Regional & Economic Indicators

- **State Location (`addr_state`):** Retained to capture state-level macro-economic shocks and legal default variations.
- **Debt-to-Income (`dti`):** Capped extreme outliers; extreme DTI is a primary driver of borrower insolvency.

Bureau Missingness Preservation

- **Binary Indicators (`*_isna`):** Created missing flags for credit bureau features.
- Absence of bureau records often carries strong risk signals (e.g., thin credit files).

Methodology & Preprocessing

Linear & Distance Models

- **Models:** Logistic Regression, Lasso, KNN, SVM.
- **Imputation:** Train median.
- **Scaling:** `StandardScaler` on 65 continuous features.
- **Encoding:** One-hot encoding (138 total features).

Tree-Based Models

- **Models:** Decision Tree, Random Forest, HistGradientBoosting.
- **Handling NaNs:** Preserved native semantic NaNs for optimal splits.
- **No Scaling:** Monotonic relationships retained.
- **Optimization:** `float32` cast for memory efficiency.

Methodology & Preprocessing

Key Observations from ROC / AUC:

- **Gradient Boosting** achieves the highest performance with an AUC of **0.709**.
- **SVM** (0.684), **Logistic Regression** (0.688), and **Random Forest** (0.681) form a strong competitive cluster.
- **Naive Bayes** shows the lowest discriminative power (AUC = 0.603).
- Confirms robust ranking across all tested algorithms on out-of-time test data.

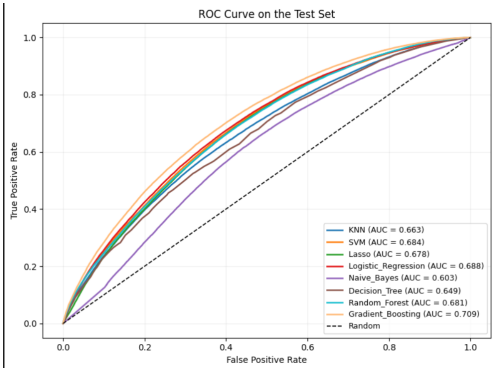


Figure: ROC Curves on the Test Set

Methodology & Preprocessing

Bootstrap Confidence Intervals:

Model	AUC (Test)	CI 2.5%	CI 97.5%
Gradient Boosting	0.7087	0.7062	0.7111
Logistic Regression	0.6885	0.6858	0.6910
Difference (GB - LR)	**0.0202**	**0.0191**	**0.0215**

Key Takeaway: The performance gain of Gradient Boosting over Logistic Regression (+0.0202 AUC) is statistically significant (confidence intervals do not overlap with zero).

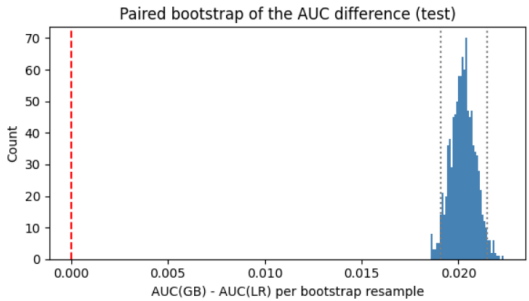


Figure: Paired Bootstrap of the AUC Difference

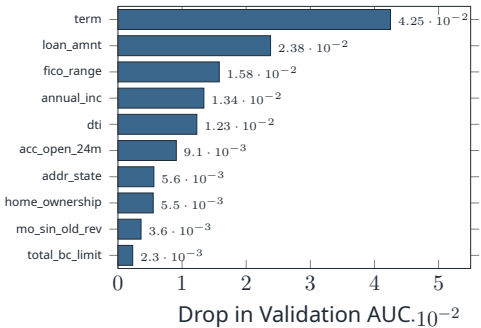
Performance Benchmarking

Performance ranking evaluated on Validation and confirmed on Test set:

Model	Valid AUC	Valid Gini	Valid KS	Test AUC	Test Gini	Test KS
HistGradientBoosting	**0.719**	**0.439**	**0.315**	**0.709**	**0.417**	**0.302**
Logistic Regression	0.703	0.406	0.293	0.688	0.377	0.275
Linear SVM	0.700	0.400	0.289	0.684	0.367	0.268
Random Forest	0.698	0.397	0.284	0.681	0.362	0.262
Lasso (L1 Regularized)	0.694	0.389	0.281	0.678	0.356	0.263
KNN ($k = 1000$)	0.678	0.356	0.255	0.663	0.325	0.233
Decision Tree	0.667	0.334	0.237	0.649	0.299	0.206
Naive Bayes	0.615	0.230	0.187	0.603	0.206	0.173

$\rightarrow$ **Champion Model:** HistGradientBoosting demonstrates superior robustness and highest test discrimination (AUC 0.709).

Performance Benchmarking



Key Risk Drivers:

1. **Loan Term (`term`):** Dominant predictor; 60-month loans substantially increase default risk.
2. **Loan Amount (`loan_amnt`):** Higher exposure raises repayment burden.
3. **FICO Score (`fico_range`):** Primary baseline creditworthiness metric.
4. **DTI & Income:** Leverage ratios critically dictate financial stability.

Risk Cut-off & Credit Scoring Metrics

Banking Credit Scoring Metrics

Evaluating credit risk requires specialized metrics beyond standard machine learning accuracy.

- **Gini Index (*Gini* $= 2 \times$ AUC $- 1$):** Measures the rank-ordering capability of the model.
- **Kolmogorov-Smirnov (KS Statistic):**

$$\text{KS} = \max_{t} |\text{TPR}(t) - \text{FPR}(t)|$$

  Quantifies peak separation power between good and bad borrower distributions.
- **Brier Score & Log-Loss:** Evaluates PD calibration accuracy (essential for Basel II capital requirements).

Risk Cut-off & Credit Scoring Metrics

Asymmetric Loss in Credit Underwriting

False Negative (Type II) loss is significantly larger than False Positive (Type I) loss.

Decision / Actual	**Actual Good ($y = 0$)**	**Actual Bad ($y = 1$)**
Approve ($\hat{y} = 0$)	+ Interest Income	**– Principal Loss (FN)**
Reject ($\hat{y} = 1$)	– Opportunity Loss (FP)	+ Loss Avoided

$\rightarrow$ **Conclusion:** Default threshold $t = 0.5$ fails because it treats errors symmetrically, leading to massive portfolio credit losses.

Risk Cut-off & Credit Scoring Metrics

Metric	Core Focus	When to Use?
Accuracy	Overall correct %	Balanced classes only
Precision	% of predicted defaults that are true	When rejection cost is high
Recall	% of actual defaults captured	When missing defaults is costly
F1-Score	Harmonic mean of Precision & Recall	Quick balanced model summary
ROC-AUC	Rank-ordering power (threshold-free)	Overall model comparison
Gini Index	Scaled AUC ($2 \times \text{AUC} - 1$)	Banking scorecard standard
KS Statistic	Peak separation distance ($\max(\text{TPR} - \text{FPR})$)	Finding optimal risk cutoff

Risk Cut-off & Credit Scoring Metrics

1. Failure of Default Threshold $t = 0.5$

- Given a $\sim$20% default rate, predicted probabilities rarely exceed 0.5.
- Approving via $t = 0.5$ accepts 97.1% of applicants but yields a severe **Recall of only 7.3%**.

2. KS Peak Optimal Threshold ($t^* = 0.166$)

- Maximizes statistical separation distance: $t^* = \arg\max_t(\text{TPR} - \text{FPR})$.
- Calibrated on Validation set and validated strictly on Test set.

Risk Cut-off & Credit Scoring Metrics

KS Peak Threshold Selection ($t^* = 0.166$):

- The chart displays TPR (Recall), FPR, and the difference (TPR - FPR) across various thresholds for Gradient Boosting.
- The optimal threshold $t^* = 0.166$ (dashed vertical line) maximizes the Kolmogorov-Smirnov (KS) statistic.
- This threshold balances loan approval rates while effectively catching high-risk defaulting borrowers.

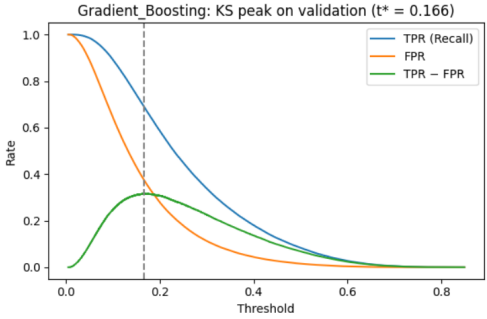


Figure: KS Peak on Validation Set

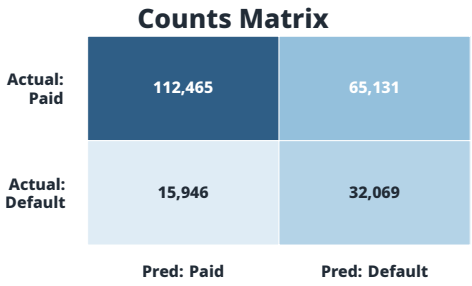
Risk Cut-off & Credit Scoring Metrics

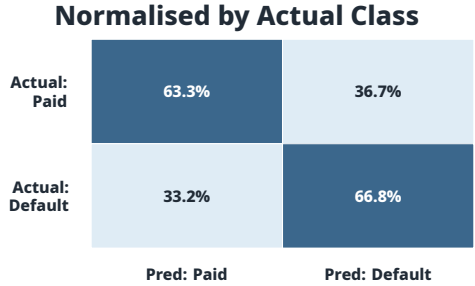
Test Set Performance Comparison ($t = 0.500$ vs. $t^* = 0.166$):

Operational Metric	Default Threshold ($t = 0.500$)	Optimal Threshold ($t^* =$
Accuracy	79.0%	64.1%
Recall (Defaults Caught)	7.3%	**66.8%**
Precision	54.4%	33.0%
Approval Rate	97.1%	56.9%
Approved Portfolio Bad Rate	**20.3%**	**12.4%**

$\rightarrow$ **Business Impact:** Threshold $t^* = 0.166$ captures **66.8%** of defaults and cuts portfolio bad rate from **21.3% down to 12.4%**.

Risk Cut-off & Credit Scoring Metrics





Key Takeaway

At optimal cutoff $t^* = 0.166$, the champion model successfully detects **66.8% (32,069)** of all defaulting loans on the out-of-time test set.

Risk Cut-off & Credit Scoring Metrics

Temporal Stability Evaluation

Formula: $\text{PSI} = \sum_{i=1}^{B} (A_i - E_i) \times \ln\left(\frac{A_i}{E_i}\right)$ (< 0.1: *Stable;* 0.1–0.25*: Moderate shift*)

- **Score Stability (Valid $\rightarrow$ Test):** PSI $= \mathbf{0.004} < 0.10$. Predicted probability distribution is exceptionally stable across 2016 to 2017–2018.
- **Feature Stability:** 56/65 features remain highly stable; 9 features (primarily credit utilization like `bc_util`) exhibit moderate drift (0.10–0.24).
- **Conclusion:** Model is safe for current deployment; monitoring required for credit utilization metrics.

Conclusions & Limitations

- **Rigorous Framework:** Processed 1.35M resolved loans with strict time-based splits and 56 leakage features removed.
- **Champion Model:** HistGradientBoosting leads with test AUC 0.709, outperforming linear baselines.
- **Optimal Cut-off:** Moving from $t = 0.5$ to $t^* = 0.166$ successfully catches 66.8% of defaults and reduces portfolio bad rate to 12.4%.
- **Deployment Ready:** Score PSI (0.004) confirms high temporal stability across economic shifts.

Conclusions & Limitations

Study Limitations

- **Selection Bias:** Unobserved rejected applicant pool.
- **Right-Censoring:** Unmatured 2018 loans understate late default risks.
- **Computational Caps:** Subsampling and hyperparameter tuning constraints.

Future Research

- **Cost-Matrix Optimization:** Integrate Loss Given Default (LGD) into thresholds.
- **LC Grade Benchmarking:** Direct comparison against proprietary platform ratings.
- **Adaptive Retraining:** Trigger automated pipelines upon severe PSI drift.

THANK YOU FOR LISTENING!

Questions & Discussion (Q & A)

Group 6 – Statistical Learning